Water Troubles Downstream: How Rivers Carry Rainfall Shocks to Conflict in India

Mahiro Kobayashi *†‡

November 5, 2026

Click here for the latest draft · Online appendix

Abstract

Most studies of rainfall and conflict assume that the only economically relevant rainfall is the one that fell locally or nearby. In a country crossed by large rivers that assumption may fail: rain that falls far upstream arrives downstream as river discharge and flooding. I study how upstream rainfall affects violent conflict downstream through river networks in India. I measure monsoon rainfall over each Indian district's entire upstream drainage area on the hydrological network, and I simulate the river discharge this rain produces with two standard hydrological models. Upstream rainfall raises conflict events by 17 percent per standard deviation the following year, while the same monsoon's local rain lowers them, so studies that use local rainfall alone mix two opposing effects. The simulated discharge intensifies conflict as well. The evidence points to the opportunity cost of fighting as the mechanism: the water arrives downstream as a hazard to farmers and lowers the return to legal work relative to fighting. The findings imply that flooding, not only drought, matters for conflict in India, and that existing policies and infrastructure may not be effective in mitigating a shock that arrives from upstream.

I. Introduction

A large empirical literature connects rainfall to violent conflict (Hsiang et al., 2013; Burke et al., 2015), and almost all of it treats a district's own rainfall as the relevant shock (Hidalgo et al., 2010; Harari and La Ferrara, 2018; Vanden Eynde, 2018; Crost et al., 2018; Fetzer, 2020; Gatti et al., 2021). The implicit assumption is that the only economically relevant rainfall is the one that fell locally. For a country crossed by large river systems that assumption may fail: the Ganges at Patna carries the monsoon of Nepal and Uttar Pradesh, not of Patna. Rain that falls upstream arrives downstream as river discharge and flooding, and no empirical study has asked how it affects violent conflict. This paper investigates the link in India.

India is an ideal setting for the question for three reasons. First, its rivers carry rain from far away: the Ganges and the Brahmaputra drain the Himalaya, and a district's upstream catchment typically spans hundreds of sub-basins and often reaches into Nepal, China, Bhutan or Pakistan, so

*Graduate School of Public Policy, University of Tokyo.

†The author would like to thank Professor Yasuyuki Sawada, Professor Yoshito Takasaki, Professor Eric Weese, and the members of the Graduate Sawada Seminar for their useful comments and advice. The views expressed in this paper, as well as any remaining errors, are the author's alone.

‡Last modified on November 5, 2026.

the water a district receives and the rain that falls on it are distinct quantities. Second, India floods often, and one season delivers the water. Over 1990–2025, EM-DAT records 247 flood disasters in India, the second-largest count of any country, with the most flood deaths and the second-most people affected (Delforge et al., 2025). 77 percent of these floods, like 78 percent of upstream rainfall, come in the June–September monsoon, which gives the shock a clear timing. Third, India combines several long-running insurgencies, with 7,564 conflict events located to districts over three and a half decades, with a rural economy in which farming is the main source of work and with district-level data on crops, wages and households, so the mechanism can be followed from the water to the labour market. Its 5,751 large dams, which can hold the water back, and its employment guarantee, which has been shown to mitigate the effect of local rainfall on conflict, offer two policy tests.

I measure monsoon rainfall over each Indian district's entire upstream drainage area, traced through 21,622 sub-basins of the hydrological network (Lehner and Grill, 2013) and weighted by how far each sub-basin's water travels along the river to reach the district. Every sub-basin that overlaps the district or its neighbours is excluded, so the treatment is rain that fell strictly elsewhere. To measure the river discharge and flooding caused by upstream rainfall, I simulate the water's path with two standard hydrological models: HBV, a land-surface model, turns the catchment's rainfall into runoff (Beck et al., 2020), and CaMa-Flood, a river-routing model, carries that runoff down the river network to the district (Yamazaki et al., 2011). I use these simulations rather than observed flood or discharge data to avoid the endogeneity of observed data: gauge records reflect dam releases and canal withdrawals, and flood catalogues suffer from reporting bias that is likely correlated with conflict (Patel, 2024).

I find evidence that upstream rainfall intensifies conflict downstream. A one-standard-deviation increase in last monsoon's upstream rainfall raises UCDP conflict events by 17 percent in the following year, while the same monsoon's rainfall on the district itself lowers conflict. The simulated discharge also raises conflict events by about 24 percent per standard deviation in the following year. The shock does not travel through trade: the rainfall of trading partners reached by road and rail, with river-linked districts excluded, does not raise conflict. I also find that the intensifying effect of upstream rainfall on conflict is concentrated in districts that depend on agriculture, the elasticity rising by 0.60 per standard deviation of dependence; it follows the harvest calendar, absent during the monsoon and the kharif harvest and rising from January of the following year; and it comes with a fall in the real agricultural wage of 0.13 percent for each 1 percent of upstream rainfall. These findings are consistent with upstream rainfall arriving downstream as a hazard to farmers and agricultural workers that lowers the opportunity cost of joining conflict. I find no evidence that either policy lever mitigates the shock: the dam contrast shows no buffering, and the employment guarantee, which mitigates the effect of local rainfall, does nothing detectable to the upstream effect.

The paper makes two contributions. First, to the best of my knowledge, it is the first empirical study to measure rainfall over a district's entire upstream drainage area using the hydrological network, and it shows that the measure matters: local and upstream rain push conflict in opposite directions, so a design built on local rainfall alone, which is what the climate–conflict literature has used (Harari and La Ferrara, 2018; Vanden Eynde, 2018; Fetzer, 2020; Gatti et al., 2021), mixes the two and may understate both. Existing work on spatially transmitted shocks follows

proximity (Harari and La Ferrara, 2018), prices (Dube and Vargas, 2013; McGuirk and Burke, 2020) or people (McGuirk and Nunn, 2025; Castells-Quintana et al., 2022; Gehring and Schaudt, 2026), not the river. The two previous studies that consider upstream rainfall, Duflo and Pande (2007) and Sarsons (2015), define upstream as the neighbouring district. For a large basin the neighbouring district holds only a small part of the water that arrives, and its rainfall is highly correlated with the district's own; my construction excludes it altogether. Second, this paper is the first empirical study to measure river discharge and flooding with an exogenous, model-simulated series in place of gauge records and flood catalogues, which are subject to endogeneity.

Two implications follow. First, flooding matters, not only drought: empirical work on rainfall and conflict has largely read rainfall shocks as income losses from too little rain, and the shock documented here is an increase in water arriving from upstream. Second, India's employment guarantee, which mitigates the effect of local rainfall on conflict (Fetzer, 2020), does nothing detectable to the upstream effect, so existing policy may prevent only part of rainfall-driven conflict. Two questions remain open, and I continue to work on both: (i) whether dams dampen or redirect the water's effect, which requires a new identification strategy to address the endogenous placement of dams, and (ii) whether the employment guarantee can reach a shock that arrives from upstream, which requires administrative programme data.

Section II describes the data. Section III presents the construction of the treatment, the specification and the identification. Section IV reports the main results. Section V examines the mechanism: the routed water (Section V.A), the trade network (Section V.B) and the opportunity-cost hypothesis (Section V.C). Section VI reports the current evidence on dams and the employment guarantee. Section VII concludes.

II. Data and Descriptive Results

A. Violent Conflict

The outcome is the district-year count of violent events from the UCDP Georeferenced Event Dataset (Sundberg and Melander, 2013), version 26.1, geocoded to 2001-census district boundaries. UCDP defines an event as the use of armed force by an organized actor against another organized actor or against civilians, causing at least one direct death at a specific location and date; events enter only if the actors belong to a conflict that has crossed twenty-five deaths in a calendar year, in one of three categories: state-based conflict, non-state conflict, and one-sided violence against civilians.[1] Three consequences of this definition matter here. First, the one-death floor means the count measures lethal incidents, not unrest generally. Second, the death must be direct, so mortality from the floods the treatment delivers — drowning, displacement, disease — does not enter the outcome. Third, the actors must be organized, so communal violence enters only as lethal two-sided clashes between communal groups (the communal category of Section IV.B). The outcome is therefore organized armed violence: insurgency (the state's fighting with armed groups and those groups' attacks on civilians) accounts for 91 percent of events, and clashes between

[1] State-based conflict: armed force between two parties, at least one the government of a state, over a contested incompatibility concerning government or territory, causing at least twenty-five battle-related deaths in a calendar year. Non-state conflict: armed force between two organized armed groups, neither the government of a state, causing at least twenty-five battle-related deaths in a year. One-sided violence: armed force by a government or formally organized group against civilians, causing at least twenty-five civilian deaths in a year; extrajudicial killings in custody are excluded. In the estimation sample, 65.0 percent of events are state-based, 4.7 percent non-state, and 30.3 percent one-sided.

communal groups for 4 percent. The remainder is border fighting between the Indian and Pakistani armies along the Line of Control and a few unaffiliated attacks.[2] Events located only to a state or coarser (10.3 percent of the 1990–2025 record) are not included in the analysis.

B. *Rainfall, Temperature, and Hydrography*

Rainfall.—Precipitation comes from CHIRPS version 3 (Funk et al., 2026), a 0.05-degree daily product that blends satellite estimates with station observations. It suits the design better than gauge-only grids, reanalysis, or interpolated climatologies: its satellite component reaches the upstream catchments in Nepal, China, Bhutan, and Pakistan that India's gauge-based grid does not cover, its resolution matches the sub-basins from which catchments are built, and it spans the whole sample. Online Appendix A.4 sets out the choice and re-estimates the headline under five alternative products.

Temperature.—The area-weighted district mean over the same window comes from CHIRTS-ERA5 (Funk et al., 2019; Verdin et al., 2020), which merges the Climate Hazards Center's station-calibrated temperature record with ERA5 daily variability on the same 0.05-degree grid as CHIRPS.

Hydrography.—The drainage network is HydroBASINS level 10 (Lehner and Grill, 2013): 21,622 sub-basins covering the Indian subcontinent and its transboundary catchments, each linked to the sub-basin it drains into. I use it to trace each district's upstream catchment and to apply the neighbour exclusion of Section III.A, which drops every sub-basin overlapping the district or a neighbouring district so that the treatment does not mix in local weather, and its coarser levels to cluster standard errors by river system (Section IV); Online Appendix A.2 explains the choice of level.

C. *Dams and Routed Water*

Dams.—Data on dams come from the National Register of Large Dams, the Central Water Commission's census of India's 5,751 large dams, where "large" follows the register's criteria: at least 15 meters high, or 10–15 meters with additional conditions on crest length, reservoir capacity, or spillway discharge. The register reports each dam's location in degrees, minutes and seconds, which are parsed directly rather than matched to an external reservoir database; 5,736 entries (99.7 percent) carry a readable coordinate, and every one falls on the HydroBASINS network, so 3,637 of the 3,638 dams commissioned by 1989 (227 billion cubic metres of gross storage) are placed on the drainage network.

Routed water.—Rainfall on an upstream catchment is a proxy for the water that actually arrives downstream, so the analysis also uses a physically routed measure of upstream water: daily river discharge simulated over 1989–2025 with CaMa-Flood (Yamazaki et al., 2011, 2013). CaMa-Flood (Catchment-based Macro-scale Floodplain model) is a global river-routing model developed in hydrology to simulate river discharge, water level and floodplain inundation from runoff. It divides the land into unit catchments along a river network derived from high-resolution topography (Yamazaki et al., 2019), gives each catchment a channel and a floodplain, and solves the flow between catchments with the local inertial form of the shallow-water equations (Bates et al., 2010; Yamazaki et al., 2013), so that water backs up, spills onto floodplains and divides

[2] Removing the India–Pakistan dyad from the outcome leaves the headline estimate at 1.46 (standard error 0.37).

where rivers split (Yamazaki et al., 2014) as it moves downstream. Its only time-varying input is runoff, which here is generated from the same CHIRPS rainfall and CHIRTS temperature that define the treatment by HBV, a conceptual land-surface model whose parameters were regionalised worldwide from observed streamflow (Beck et al., 2020); the model runs at 0.1-degree (roughly 10-kilometre) resolution over South Asia. The series is summarised as the June–September discharge total at the district's river, entered as a within-district anomaly. Section V.A describes the measure and its construction; Online Appendix C documents construction and validation in detail.

D. Sample Construction and Summary Statistics

The main estimation sample is fixed by two restrictions. First, of the 593 districts in the 2001-census boundary file, 168 have no upstream catchment left after the neighbour exclusion — both defined in Section III.A — and so receive no treatment. Second, of the remaining 425, 215 record no conflict event between 1990 and 2025. Since the fixed-effects Poisson, the estimation method used in this paper, conditions on each district's total count, these all-zero districts carry no information about the coefficient and drop out by construction. The headline estimates therefore use 210 districts and 7,560 district-years.

Table 1 reports the summary statistics of the headline variables. With district fixed effects, the identifying variation is a standard deviation of about 172 mm around each district's own mean, against 351 mm in the raw cross-section. Panel B separates the monsoon from the rest of the year: upstream catchments receive 1,002 mm between June and September against 290 mm in the other eight months, and the monsoon carries almost twice the within-district variation (172 mm against 89 mm). Online Appendix Figure E.4 shows that 78 percent of upstream rainfall falls in the monsoon, with nearly the same monthly profile as the district's own rainfall. A district-year experiences 1.001 events on average, with a median of zero and a maximum of 197; roughly four in five district-years record none (incidence 0.17). Unlike rainfall, conflict is not concentrated in the monsoon: only 36 percent of events fall between June and September. Panel C's routed-water measures cover fewer district-years (6,336).[3] They are far more concentrated than rainfall, too: in an average year a district's river exceeds its two-year level — the flow its annual peak reaches about one year in two — on 2.3 days and its five-year level on 0.6, and in 79 percent of district-years it never reaches the two-year level.

Figure 1 maps the three objects. Conflict is spatially concentrated — a Maoist belt through central and eastern India, the Kashmir valley, and the Northeast — while upstream rainfall is high along the Himalayan front and the western coast and low across the northwest. Panel C shows the districts in the estimation sample.

[3]Routed water is defined for 176 of the 210 estimation districts. Of the rest, 15 contain no river channel the routing model resolves, 3 are headwater districts left with almost no upstream input once the 50-kilometre mask removes their nearby runoff, and 16 lie below a river split, where removing nearby runoff shifts how the flow divides between branches, so the routed measure is not reliable there.

Table 1: Summary statistics, estimation sample

	N	Mean	SD	Within-district SD	Min	Max
A. Conflict						
Conflict events	7,560	1.001	5.638	1.729	0.000	197.000
Conflict incidence (0/1)	7,560	0.173	0.378	0.299	0.000	1.000
B. Rainfall and temperature						
Monsoon (June–September)						
Upstream rainfall (mm)	7,560	1,001.776	350.989	172.122	62.793	3,182.764
Local rainfall (mm)	7,560	1,054.856	512.680	200.528	21.175	4,323.204
Upstream temperature (°C)	7,560	29.328	4.536	0.552	13.143	36.962
Local temperature (°C)	7,560	32.862	2.711	0.569	19.126	39.017
January–May and October–December						
Upstream rainfall (mm)	7,560	289.816	211.082	88.948	6.763	1,604.946
Local rainfall (mm)	7,560	334.024	317.905	109.456	1.476	2,208.725
Upstream temperature (°C)	7,560	26.043	6.987	0.543	3.247	34.902
Local temperature (°C)	7,560	29.937	4.084	0.552	4.119	36.404
C. Routed water (CaMa-Flood)						
Upstream discharge, June–September (log)	6,120	11.581	2.093	0.379	4.169	15.708
Upstream flood-stage days, 2-year level	6,336	2.324	7.209	3.818	0.000	75.000
Upstream flood-stage days, 5-year level	6,336	0.574	2.945	1.502	0.000	44.000
Peak discharge / 2-year level	6,336	0.744	0.489	0.306	0.000	9.700

Notes: District-years in the Poisson estimation sample: the 210 districts with at least one located event, 1990–2025. The non-monsoon block covers January–May and October–December. Within-district SD is the mean of the districts' within-district standard deviations. Upstream rainfall is the river-distance-weighted total over the catchment after the neighbour exclusion (Section III). Panel C covers districts with a CaMa-Flood river cell and an upstream catchment; upstream discharge is the June–September total of upstream-origin discharge at the district's river cell, the measure of Section V.A; a flood-stage day is a day whose centred three-day mean discharge exceeds the cell's return level.

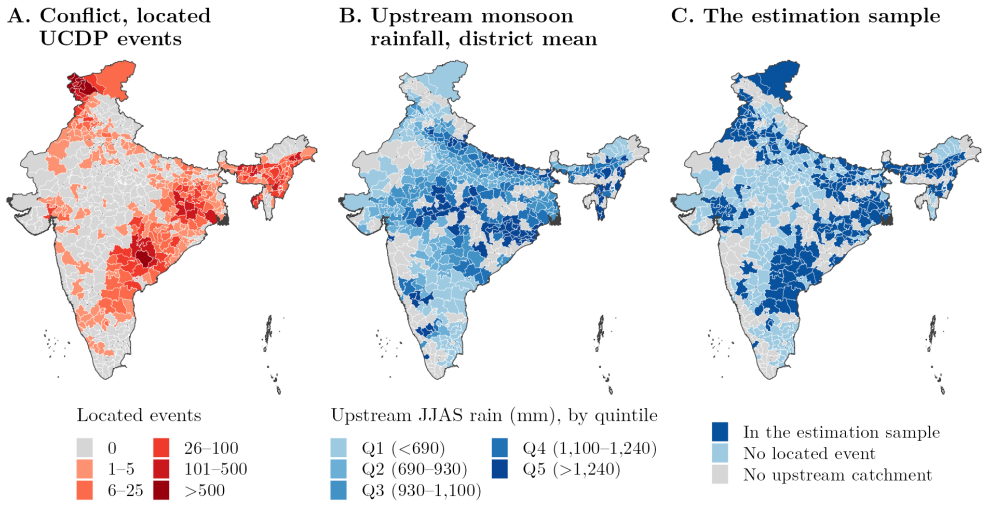


Figure 1: Conflict, upstream rainfall, and the estimation sample

Notes: Panels A and B cover all 593 mapped districts. Panel A: located UCDP events, 1990–2025. Panel B: mean June–September upstream catchment rainfall, in quintiles. Panel C: the 210 districts in the estimation sample (7,564 located events); of the rest, 168 have no upstream catchment and the others record no located event, so the Poisson drops them.

III. Empirical Strategy

A. *Constructing Rainfall from Elsewhere*

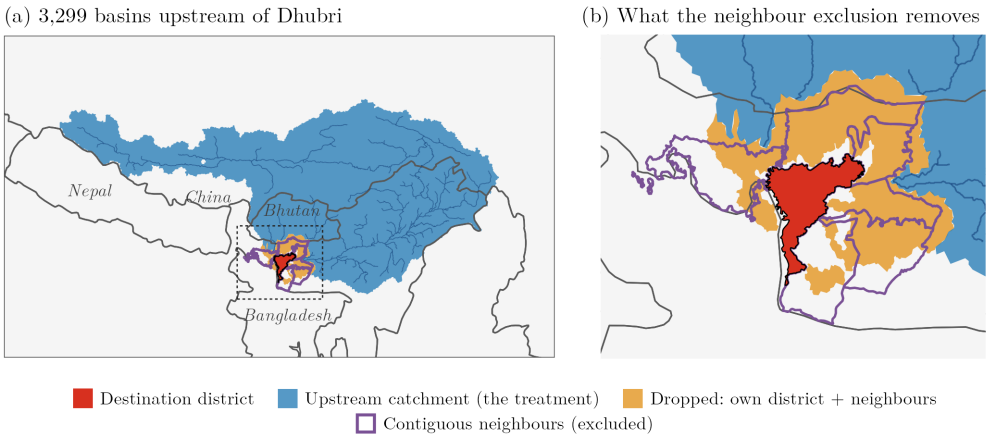


Figure 2: The upstream treatment, for one district

Notes: Panel (a): the 3,299 sub-basins draining into Dhubri (Assam), the district at the 90th percentile of the estimation sample by number of upstream sub-basins. Panel (b) enlarges the dashed box. Sub-basins overlapping the district, its 6 contiguous neighbours (purple outline) or adjacent foreign units are excluded (amber, 111 sub-basins). The treatment is monsoon rainfall over the remaining (blue) sub-basins, weighted by inverse river distance to the district.

For district i, let $U(i)$ denote the set of sub-basins[4] that drain into i through the hydrological network, *excluding* every sub-basin that overlaps i itself or any district sharing a border with i. Across an international border, the adjacent administrative unit of the neighbouring country counts as a bordering district. This neighbour exclusion is the preferred choice, and Figure 2 visualizes what it does: it removes the district's own weather and its immediate neighbourhood's weather from the treatment, so that identifying variation comes from rain the district never saw. Upstream rainfall is a weighted average in which each sub-basin counts by how far its water must travel along the river:

$$R_{it}^{UP} = \sum_{b \in U(i)} w_{ib}\, R_{bt}, \qquad w_{ib} = \frac{\omega_{ib}}{\sum_{b' \in U(i)} \omega_{ib'}}, \qquad \omega_{ib} = \sum_{p \in P(i,b)} \frac{s_p}{1 + r_p},$$

where R_{bt} is sub-basin b's monsoon rainfall, $P(i,b)$ is the set of river paths from sub-basin b to district i, r_p is the length in kilometres of path p along the network, from sub-basin b's outlet to where its water enters district i, and s_p is the share of b's flow that follows p (Online Appendix Figure A.4 draws each piece of this notation). On a single channel there is one path and $s_p = 1$, so $\omega_{ib} = (1 + r_{ib})^{-1}$; where a river splits, the branch carries the share of the flow it receives in a hydrological routing model (Online Appendix A.3). The weight therefore decays with distance along the river, and Online Appendix A.3 shows that the estimates are not sensitive to this choice.

B. *Specification*

The estimating equation is a fixed-effects Poisson regression,

$$\mathbb{E}[A_{it} \mid \cdot] = \exp\Big(\beta_1 \log R_{it}^{UP} + \beta_2 \log R_{i,t-1}^{UP} + \mathbf{u}_{it}'\boldsymbol{\gamma} + \mathbf{x}_{it}'\boldsymbol{\theta} + \rho\, \bar{A}_{i,t-1} + \bar{\mathbf{x}}_{it}'\boldsymbol{\lambda} + \alpha_i + \delta_t\Big), \qquad (1)$$

[4] HydroBASINS level 10 throughout; Online Appendix A.2 explains the choice of level.

where A_{it} is the district-year count of UCDP events, $\mathbf{u}_{it}$ is current and lagged upstream temperature, $\mathbf{x}_{it}$ is current and lagged local rainfall and temperature, and α_i, δ_t are district and year effects. A bar denotes a neighbourhood average: for any district-level variable v, $\bar{v}_{it} = \sum_{j \in N(i)} W_{ij} v_{jt}$, where $N(i)$ is the set of districts sharing a border with i and the weights W_{ij} are row-normalised so that they sum to one.[5] Thus $\bar{A}_{i,t-1}$ is neighbours' lagged conflict, entered as $\log(1+\text{count})$, and $\bar{\mathbf{x}}_{it}$ is neighbours' local weather.

Equation (1) is a variant of the spatial Durbin model that Harari and La Ferrara (2018) use for conflict: besides the district's own regressors, it includes its neighbours' conflict and weather, so that violence spreading across district borders and weather shocks common to a region are not attributed to upstream water. Equation (1) omits three terms of the preferred model of Harari and La Ferrara (2018). The district's own conflict in the previous year and neighbours' conflict in the same year cannot be estimated consistently in this setting: the first is entangled with the district fixed effect in a fixed-effects Poisson model, and the second is determined jointly with the outcome. Neighbours' conflict therefore enters with a one-year lag, $\bar{A}_{i,t-1}$. Neighbours' upstream rainfall is left out since neighbouring districts drain overlapping catchments, which makes it nearly a copy of the district's own (correlation 0.91 after fixed effects). Online Appendix B.3 adds the three terms back, one at a time and all together, to check that these omissions do not drive the result.

Local weather is essential here to isolate the effect of upstream rainfall. Upstream and local monsoon rainfall correlate at 0.71 after fixed effects, because the same weather system waters a district and its catchment. They act through different channels: local rain directly, on the district's own crops and roads; upstream rain plausibly through the river. Holding local weather fixed isolates the transmitted channel through the river.

C. Identification and Inference

Upstream rainfall is a shift–share,

$$R^{UP}_{it} = \sum_{b \in U(i)} w_{ib}\, R_{bt}, \qquad \sum_{b} w_{ib} = 1,$$

with exposure shares w_{ib} fixed by hydrography and shocks R_{bt} the basin-year rainfall. Identification rests on the exogeneity of the shocks, not the shares (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022). Let μ_i be district i's expected exposure, the treatment it would receive on average over draws of basin rainfall (Borusyak and Hull, 2023). The identifying assumption is

$$\mathbb{E}\big[(\log R^{UP}_{it} - \mu_i)\, \varepsilon_{it}\big] = 0 \tag{2}$$

after district and year effects and local weather are partialled out: whether a district's catchment receives more or less rain than usual in a given year is unrelated to the other, unobserved causes of conflict in that district that year. The treatment is built from basin rainfall, so this is in substance a condition on the shocks: rainfall above or below a basin's climatology must be unrelated to the unobservables of the districts it drains (Borusyak et al., 2022). It is an assumption about

[5] So $\bar{v}_{it}$ is simply the average of v across district i's immediate neighbours in year t, and district i itself is never among them: $W_{ii} = 0$. Row-normalising means a district with three neighbours and one with eight are each compared with the average of their own, rather than the district with more neighbours receiving a mechanically larger term.

weather, not about where rivers run, and it identifies from deviations from climatology in the manner standard to the climate–economy literature (Dell et al., 2014).

Two things follow. First, expected exposure must not bias the estimate. When climatology is stable, μ_i is constant and district effects absorb it; the demanding test lets climatology drift and recenters on a decade-specific μ_i (Borusyak and Hull, 2023). This is also the margin on which Christian and Barrett (2024) caution that weather–conflict panels can manufacture correlations out of slow-moving common components. Second, the treatment's variation comes from basin rainfall, so the estimate is consistent as the number of independent basin shocks grows, not the number of districts (Borusyak et al., 2022). Basin rainfall is spatially correlated within a season, so the independent shocks are far fewer than the districts, and treating districts as independent would overstate precision. Standard errors therefore cluster on whole river systems and treat basin shocks as the sampling unit (Adão et al., 2019), and randomization tests redraw the shocks themselves. Section IV and Online Appendices B.4 and B.5 report both.

IV. Upstream Rainfall and Violent Conflict

I provide evidence that, in India, upstream rainfall raises violent conflict in downstream districts the following year. I first present the main estimates. The conservative one, without spatial controls, is that a 1 percent wetter upstream monsoon raises the expected number of conflict events the following year by about 0.91 percent. I then break the outcome down by type of violence. Lastly, I conduct sensitivity analyses. The estimate is robust to the design choices and passes the shift-share diagnostics, and what it captures is rainfall on the catchment that drains to the district rather than rainfall on land that no river connects to it.

A. Main Results

Table 2 reports the main results. Column 1 estimates equation (1) without the spatial controls and is the conservative estimate. The coefficient on last year's upstream monsoon rainfall is 0.91 (Conley standard error 0.36). It is an elasticity: a 1 percent wetter upstream monsoon raises expected conflict events the following year by about 0.91 percent, and a 10 percent wetter one by about 9.1 percent. A typical shock, one standard deviation of the identifying variation or a monsoon 19 percent wetter, raises them by about 17 percent, against a mean of 1.00 events per district-year.[6] Contemporaneous upstream rainfall is small and insignificant. Local rainfall in the same regression enters negatively: rain measured upstream and rain measured where the violence occurs move conflict in opposite directions, so regressions using local rainfall alone conflate two channels.

Columns 2–4 add spatial controls. Holding neighbours' lagged conflict fixed moves the elasticity to 0.99; holding neighbours' own weather fixed moves it to 1.35. Neighbours' weather belongs in the specification because the monsoon is spatially smooth: a district's upstream exposure still correlates 0.75 with its neighbours' local rainfall after every basin touching them is excluded, and neighbours' rain carries the pacifying local-income sign, so omitting it biases the estimate toward zero. That smoothness — rainfall changing little from one district to the next — also makes local and neighbours' rainfall nearly collinear once both enter, so from column 3 onward local rainfall

[6] In a Poisson regression on log rainfall, the coefficient is the elasticity of expected events with respect to rainfall. The effect of a one-standard-deviation shock is $\exp(0.91 \times 0.17) - 1$, where 0.17 is that standard deviation in log points.

Table 2: Upstream rainfall and conflict

	No spatial controls	Spatial controls			
	(1)	(2)	(3)	(4)	(5)
log Upstream Precip. (t)	-0.045	-0.106	-0.047	-0.086	-0.275
	(0.371)	(0.412)	(0.381)	(0.487)	(0.468)
log Upstream Precip. $(t-1)$	0.910**	0.994***	1.351***	1.460***	1.242***
	(0.359)	(0.314)	(0.351)	(0.283)	(0.277)
log Local Precip. (t)	-0.353	-0.385	-0.103	-0.088	-0.675
	(0.258)	(0.305)	(0.482)	(0.579)	(0.658)
log Local Precip. $(t-1)$	-0.921***	-0.978***	0.527	0.507	0.567
	(0.324)	(0.330)	(0.921)	(1.025)	(1.012)
log Neighbours' Conflict $(t-1)$		1.002***		0.994***	0.957***
		(0.149)		(0.149)	(0.155)
log Neighbours' Local Precip. (t)			-0.188	-0.315	1.701*
			(0.615)	(0.871)	(1.009)
log Neighbours' Local Precip. $(t-1)$			-2.046*	-2.103	-1.443
			(1.190)	(1.316)	(1.412)
log Downstream Precip. (t)					-1.329***
					(0.456)
log Downstream Precip. $(t-1)$					-0.906***
					(0.336)
1 SD of identifying variation: a wetter monsoon of (%)	19	19	12	12	12
Implied effect of a 1 SD wetter monsoon (%)	17	19	17	18	15
Observations	7,560	7,560	7,560	7,560	6,372
Number of Districts	210	210	210	210	177
Mean of dependent variable	1.00	1.00	1.00	1.00	1.06
Estimation	Poisson	Poisson	Poisson	Poisson	Poisson
District & Year FE	✓	✓	✓	✓	✓

Notes: Poisson regressions of UCDP events located to a district or better, 1990–2025. Each column includes current and lagged upstream and local temperature. Column 2 adds neighbours' lagged conflict, column 3 neighbours' current and lagged rainfall and temperature, and column 4 both (equation (1)); neighbours' variables are averages over contiguous districts, conflict as $\log(1+\text{count})$. Column 5 adds downstream weather, built the same way along the flow network in the opposite direction, on the districts where it is defined; on that sample column 4 gives 1.36. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

serves only as a control and its coefficient is not separately interpretable. Column 4 adds both and is equation (1) itself: the elasticity is 1.46 (Conley 0.28). Every column is robust to replacing the year effects with state × year effects, which absorb anything common to a state in a year: the elasticity is then 1.39 to 1.62, significant at the 1 percent level throughout (Online Appendix Table E.1).

Column 5 adds downstream rainfall to equation (1), built identically but following the flow network in the opposite direction. Its coefficients are negative, and the upstream estimate is unchanged on that sample. The downstream term is not the clean zero a placebo would ideally return; Online Appendix E examines it and Section VII discusses what it means.

B. Types of Conflicts

Table 2 pools all types of conflict, and Figure 3 estimates equation (1) on each type. Insurgency, which accounts for 91 percent of events, responds at 1.35, close to the headline; within it, battles between the state and armed groups respond at 1.35 and insurgents' attacks on civilians at 0.92, which is not significant. Communal clashes respond at 1.87, on 66 districts and too few events to be significant. By conflict system every estimate is positive, but each system is concentrated in one region, so its estimate rests on few districts (20 for Kashmir, 34 for the Northeast and 93 for Maoist violence, against 210 for all conflict) and is imprecise; over the full window only the Northeast is significant at five percent (Online Appendix E.2).

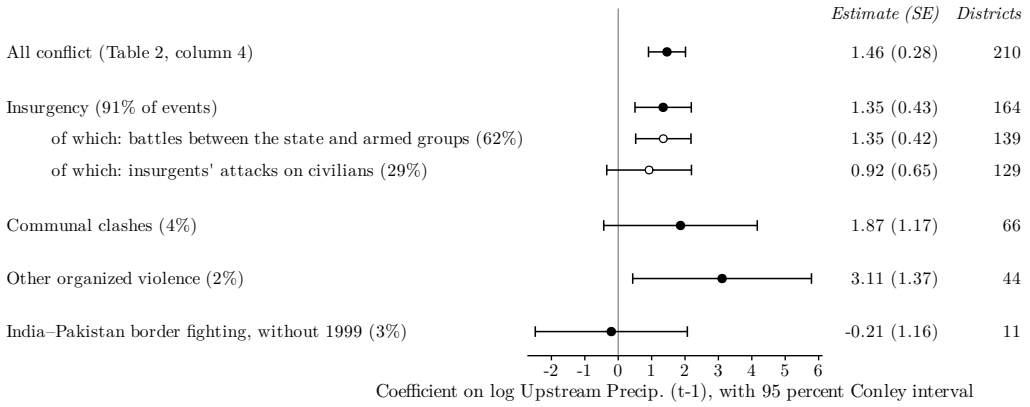


Figure 3: Effects by types of conflicts

Notes: Each row estimates equation (1) with one type of violence as the outcome, neighbours' conflict built from the same type. Markers are the coefficient on log Upstream Precip. $(t-1)$ with 95 percent Conley (150 km) intervals. Shares are percentages of located events in the estimation sample, classified by actor (Online Appendix Table A.1): insurgency is fighting between the Government of India and an armed group, or an armed group's attacks on civilians; communal is non-state violence between communal groups; other organized violence is one-sided violence by governments and fights between insurgent factions. India–Pakistan border fighting is estimated without 1999; with it the estimate is 2.86 (0.65), carried by the Kargil war.

C. Sensitivity Analysis

In this section, I (i) provide evidence that the result is not driven by the design choices, (ii) argue that the monsoon months (June–September) are the right window over which to measure rainfall, (iii) check the identifying assumption, and (iv) provide further evidence for the interpretation of the result.

Design choices.—The treatment rests on four choices: where the boundary of the upstream catchment is drawn, how its sub-basins are weighted, how finely rainfall is measured, and which years the sample covers. The estimate does not depend on any of them. The first choice is the boundary. The preferred exclusion rule drops the sub-basins of the district's own territory and of its neighbours; Online Appendix Figure A.3 replaces it with a distance-based buffer radius of 0 to 100 kilometres. The lagged upstream elasticity is positive at every radius, with estimates ranging from 0.52 to 2.28 and 37 of 44 clearing the Conley 95 percent interval, and it declines as the radius widens, as it should if the shock travels by water and attenuates with distance. The second is the weighting. Under five alternative weightings of the sub-basins, including straight-line distance and runoff volume, the estimate stays between 1.41 and 1.54 and significant at the one percent level (Online Appendix A.3). The third is the rainfall data. Averaging upstream rainfall over coarser sub-basins does not change the results qualitatively (Online Appendix Figure A.2); the sign of the effect does not depend on the rainfall product, and the effect is as large where rain gauges are scarcest (Online Appendix A.4). The fourth is the sample period. The estimate is positive in every 11-year window of the sample (Online Appendix Figure E.6).

The seasonal window.—I argue that the monsoon months (June-September) are the right window over which to measure upstream rainfall, for two reasons. The first is that they carry the water: June to September delivers 78 percent of upstream rainfall, and 80 percent of the floods the Global Flood Database records in India begin in those months. The second is that rainfall in the other months adds nothing. Table E.3 weights rainfall by each district's own crop calendar instead of the monsoon, and the elasticity falls to 0.75 against 1.46, since the added months dilute the

monsoon signal. Rainfall outside the monsoon correlates only 0.07 with monsoon rainfall and has no significant effect, and the monsoon coefficient is unchanged when it is added. The response is to the monsoon's water, not to the crop's growing season; Section V.C traces it month by month.

Shift-share diagnostics.—I conduct robustness checks addressing the concerns about shift-share identification. The first is non-random exposure: a district with a large, wet catchment receives more upstream rain in every year, and if such districts differ in conflict for other reasons the treatment is confounded. Recentering the treatment on expected exposure, estimated by decade so that climatology may drift, leaves the elasticity at 1.390 (Online Appendix B.1). The second is a small number of shocks: consistency requires many independent basin shocks, and sub-basins in one river system share a monsoon. The treatment draws on 37.4 effective shocks across the 82 major sub-catchments, above the twenty equal-sized shocks at which the simulations of Borusyak et al. (2022) find the asymptotic approximation adequate (Online Appendix B.2). The third is a single theatre carrying the result. Dropping one state at a time keeps the elasticity between 1.32 and 1.57 (Online Appendix Figure E.11). The fourth is dependence across districts: districts along one river share upstream basins, so their treatments are correlated however far apart they lie, which standard errors based on distance, such as the Conley errors of Table 2, do not capture (Adão et al., 2019). Under eight dependence structures, including exposure-robust standard errors and clustering on whole river systems, the estimate clears one percent, and the least favourable still places it more than three standard errors from zero (Online Appendix B.4). The last is coincidence in time: a trend or cycle shared by rainfall and conflict could produce the estimate without any link between the two. Re-dating the shocks breaks the link ($p = 0.0002$; Online Appendix B.5).

The catchment's water.—I address three concerns about what upstream rainfall captures. The first is that the particular map of catchments does not matter, so that any plausible assignment of upstream basins would produce the estimate. Reassigning catchments among the districts of a state never reaches the actual estimate ($p = 0.0002$; Online Appendix B.5). The second is that upstream rainfall stands for weather the district experiences itself or shares with its surroundings. Online Appendix B.6 adds the candidates to the regression one at a time: extreme local rainfall, measured as days above 50 millimetres and the wettest five-day total, which the seasonal local total in equation (1) does not capture; rainfall on the basins the neighbour exclusion removes; and the common component of the monsoon regime. Adding any of them as a control leaves the upstream coefficient unchanged in size and significance. The third is that upstream rainfall stands for weather moving along the same corridor, on land at the same distance and bearing that no river connects to the district. On seasonal totals such rainfall is too collinear with the treatment to separate (correlation 0.95 after fixed effects): entered together, upstream rainfall keeps its coefficient and the twin's is near zero. On flood-type statistics the two separate, the catchment keeping 0.078 while the ridge beside it falls to 0.020 ($p = 0.03$).

V. Mechanism

A. Routed Water

In Section IV, I provided evidence that upstream rainfall raises conflict downstream. I now test the following hypothesis: if the river is the channel that carries the shock, then the district's river discharge should predict conflict as well.

To measure discharge I first generate daily runoff from the treatment's own inputs — CHIRPS

rainfall and CHIRTS temperature — with HBV, a conceptual land-surface model whose parameters Beck et al. (2020) regionalised at 0.1-degree resolution from 4,229 gauged catchments worldwide. Next I route the generated runoff through CaMa-Flood (Section II.C). Together the two reproduce the chain from rain falling on an upstream catchment to the river rising downstream, which is illustrated in Figure 4. A weighted sum of upstream rainfall captures neither of the two steps in that chain. On the land, only part of the rain becomes runoff: the rest evaporates, is taken up by vegetation, soaks into the soil or is held as snow, in shares that depend on how wet and how cold the catchment already is. HBV simulates this step: it turns each day's rain and temperature into soil moisture, snowpack and evaporation, and into the runoff that leaves the land (steps 1–2 in Figure 4). In the river, the runoff then travels downstream. Channels and floodplains store it, tributaries add to it, and when a channel overflows the excess spreads across the floodplain, so the discharge that reaches a district depends on when and along which channels the water travels, not only on how much rain fell upstream. I use the CaMa-Flood model to simulate this step: CaMa-Flood delivers the runoff to the river network and moves it downstream (steps 3–4). The output is daily discharge at every river cell; Online Appendix C.1 gives further details.

Two features of the measure matter for identification. First, the only input that changes over time is the weather. The river map, channel dimensions and physics are fixed, no observed river data enter the model, and dams, barrages and canals are absent. Therefore, conditional on district and year effects, the routed water is plausibly exogenous. This is the reason why I rely on the model rather than on observed discharge or flood data. Gauge discharge records are sparse, and the flow they record reflects dam releases and canal withdrawals that can respond endogenously to local conditions. Flood catalogues have a different problem. The two most widely used, the Dartmouth Flood Observatory archive (Brakenridge, 2023) and EM-DAT (Delforge et al., 2025), record a flood when news articles or government reports do; they lack a consistent definition of a flood and record fewer floods where incomes are lower (Patel, 2024). The Global Flood Database (Tellman et al., 2021) maps the Dartmouth events from satellite imagery and inherits their coverage. A flood measure whose detection depends on reporting and income will introduce omitted variable bias. Second, the routing takes all runoff, the district's own included; the district's own rainfall and temperature enter every regression as controls, as they do beside the rainfall treatment, so the comparison between the two holds local weather fixed. The regressor is the June–September total of the simulated discharge at the district's largest-drainage river cell, entered, like the rainfall beside it, as a within-district anomaly. [**REVISIT:** update after the new CaMa run] Online Appendix C documents the routing and reports the earlier ERA5-Land-forced runs, which mask the district's own and nearby runoff to isolate upstream-origin water.

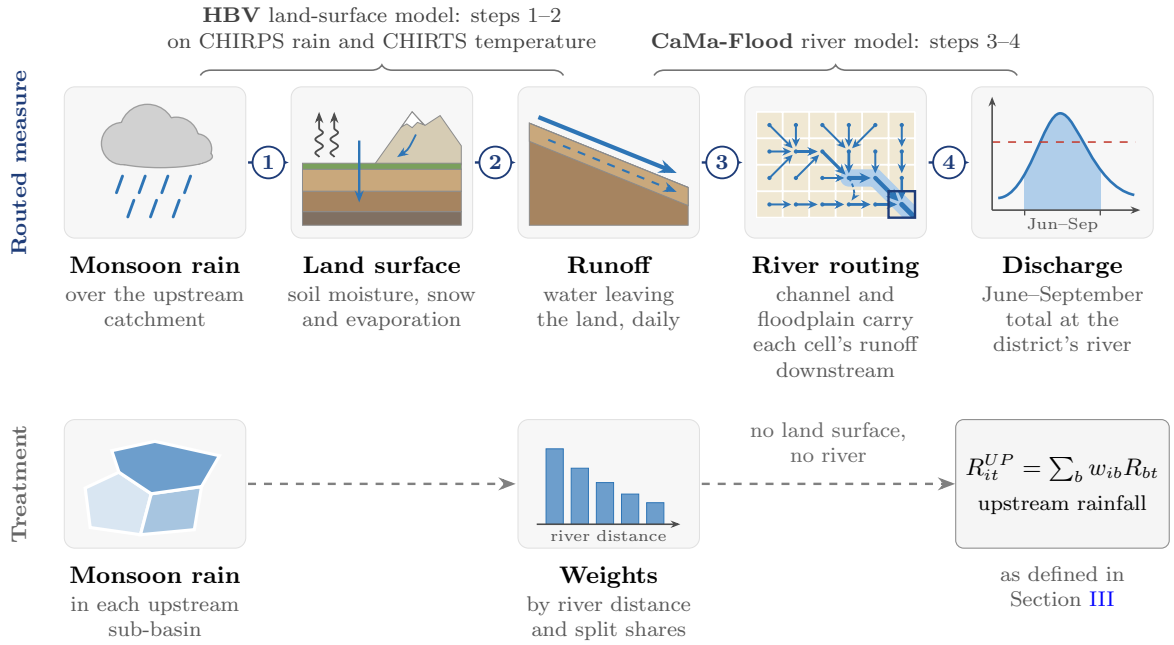


Figure 4: From monsoon rain to routed discharge

Notes: Top row: the routed measure. HBV, driven by CHIRPS rainfall and CHIRTS temperature, converts weather into soil moisture, snowpack and evaporation (1) and daily runoff (2); CaMa-Flood routes the runoff through the river network (3) to daily discharge at the district's river cell, whose June–September total, as a within-district anomaly, is the routed measure (4). Bottom row: the treatment of Section III, a weighted average of sub-basin rainfall.

Table 3 reports the results for the routed water. Rainfall and discharge enter as within-district anomalies, so a coefficient is the response to a one-standard-deviation local shock and the two regressors read on one scale. Last monsoon's discharge raises conflict when it stands alone (column 2): a one-standard-deviation rise in routed discharge raises the expected number of events in the following year by about 24 percent (0.22 log points). The river tracks the rain closely: the within-district R^2 of the discharge anomaly on the rainfall anomaly is 0.76.

One pattern stands out. Entered together (column 3), the rainfall effect does not attenuate while the discharge effect does; if discharge were the channel through which rainfall affects conflict, it is the rainfall that would attenuate. Since the river is simulated from the treatment's own rainfall, this is not a mismatch of rainfall products, and it survives state × year effects (Online Appendix Table E.2). Two interpretations remain. Either the response runs through the upstream land and economy rather than through the river, or the simulated river measures the water that reaches the district with more noise than the rainfall that made it. Section V.B tests the first through the transport network and rejects it: the shock does not arrive through trade. One possible reason for the river's null is therefore noise in the simulated variable, and Section V.C proceeds treating upstream rainfall as the measure of the water that reaches the district.

B. Shocks through market or river?

If a district's trading partners lie upstream, a wet upstream harvest could reach the district through prices and demand rather than through the river. Figure 5 tests this by swapping the river for the transport network: it builds a second rainfall measure over the districts connected to each other by road and rail. Panel (a) illustrates the trade measure over today's road network for one district. The measure does not observe actual trade. It treats every other district as a possible trading partner and weights its monsoon rainfall by road travel time, so that nearby districts count

Table 3: Upstream rainfall and the water in the river

	Rainfall (1)	River (2)	Both (3)
Upstream rainfall anomaly (t)	0.110		0.251*
	(0.095)		(0.136)
Upstream rainfall anomaly ($t-1$)	0.373***		0.418***
	(0.072)		(0.101)
Routed discharge anomaly, June–September (t)		-0.024	-0.208*
		(0.078)	(0.122)
Routed discharge anomaly, June–September ($t-1$)		0.219***	-0.048
		(0.058)	(0.074)
First link: within-district R^2 of river on rainfall		0.76	
Observations	6,336	6,336	6,336
Number of Districts	176	176	176
Mean of dependent variable	1.10	1.10	1.10
Estimation	Poisson	Poisson	Poisson
District & Year FE	✓	✓	✓
Spatial block	✓	✓	✓

Notes: Poisson regressions of UCDP events located to a district or better, 1990–2025, on one sample: districts with a CaMa-Flood river cell, an upstream signal, no bifurcation flag and a finite June–September flow in both years. Routed discharge is the June–September total of daily discharge at the district's largest-drainage river cell, simulated by CaMa-Flood from runoff that a land-surface model (HBV) generates from the same CHIRPS rainfall and CHIRTS temperature that define the treatment (Section V.A). Both regressors are within-district anomalies, $(x - \bar{x}_i)/\mathrm{sd}_i$, so a coefficient is the effect of a one-standard-deviation local shock. The first-link row is the within-district R^2 of the river anomaly on the rainfall anomaly and the controls. Every column includes the controls and the spatial block of Table 2. Conley spatial-HAC standard errors (150 km cutoff) in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

for most of the measure and distant ones for little.[7] The district's contiguous neighbours and the districts on the same river system[8] are left out, so the partners that remain are reached by road, not by river. Panel (b) compares this measure with the treatment across three road networks and the railway. The coefficient on the partners' rainfall is negative for every road and rail network, while the upstream coefficient stays between 1.387 and 1.483, compared with 1.460 without the partners' measure. The result is robust to how the partners and their weights are defined (Online Appendix Table B.5). A second test uses the direction of travel. Goods move along a road in both directions, but water moves down a river in one direction, so under a market channel the rainfall of partners at a given travel time would matter for upstream and downstream partners alike, and under a river channel only for upstream ones. At equal road travel time, upstream rain raises conflict in every band of travel time and downstream rain never does, with the pooled difference significant ($p < 0.001$; Online Appendix Figure B.5). The effect of upstream rainfall in Table 2 is therefore unlikely to reflect shocks arriving through trade networks.

C. Opportunity Cost Hypothesis

The opportunity-cost account of conflict holds that violence rises when the return to legal work falls relative to fighting (Dal Bó and Dal Bó, 2011). In rural India, farming is the main source of legal work, so a destroyed harvest is a fall in that return. I test two implications of this account: whether the response is concentrated where households depend on agriculture, and whether it arrives on the farming calendar, after the lost harvest would have been sold.

Table 4 shows that the response is concentrated in districts that depend on agriculture. Agricul-

[7] Travel time is computed between district centroids over the OpenStreetMap road network (motorway, trunk, primary and secondary roads) at class-based speeds of 80, 65, 55 and 40 kilometres per hour; Online Appendix B.6 describes the networks.

[8] A district counts as on the same river system when at least 5 percent of its area lies in sub-basins upstream of the destination district, downstream of it, or overlapping it.

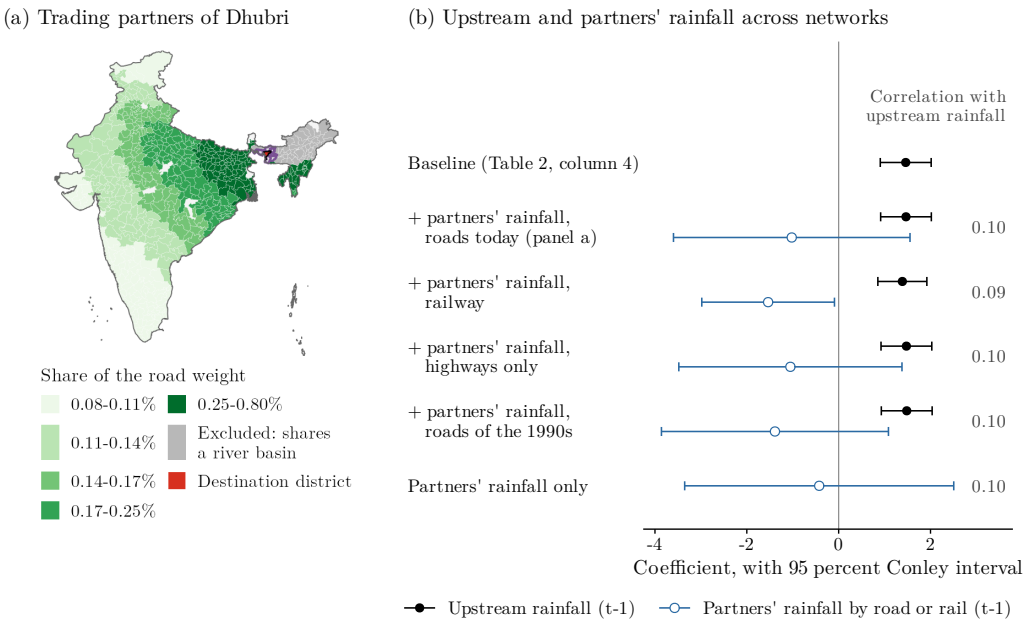


Figure 5: Rainfall through the road network and through the river

Notes: Panel (a): trading partners of Dhubri (Assam), each weighted by $1/(1+h)$, where h is road travel time in hours over today's OpenStreetMap network; excluded (grey) are Dhubri's contiguous neighbours and the districts with at least 5 percent of their area in its upstream, downstream or own river basins. Panel (b): coefficients on lagged upstream and partners' rainfall in equation (1), with 95 percent Conley (150 km) intervals. The other networks are OpenStreetMap trunk and primary roads, the Digital Chart of the World (early 1990s), and OpenStreetMap mainline railway at 40 kilometres per hour. Online Appendix Table B.5 reports the regressions.

tural dependence is measured from the 2005 Economic Census as the negative of non-agricultural jobs per resident, standardised across districts, so that a higher value means fewer jobs outside farming. Column 1 shows that a 1 percent increase in last monsoon's upstream rainfall raises expected conflict events by 1.12 percent at mean dependence, and that the elasticity rises by 0.60 per standard deviation of dependence; the result survives controlling for the interactions of rainfall with irrigation, literacy and worker density (column 2). The same 1 percent of upstream rainfall lowers the real agricultural wage by 0.13 percent, and the fall is larger where dependence is higher (columns 3–4): the districts whose violence responds most are those whose labour market is hit hardest. The interaction of lagged upstream rainfall with dependence in column 1 holds when the conflict sample is restricted to the 138 districts with a wage series: 1.03 (s.e. 0.40). Panel B adds the size of the river entering the district. The dependence gradient for conflict is steeper where the river is larger — the elasticity rises by a further 0.37 per standard deviation of dependence for each standard deviation of river size (column 1; 0.38 with the rival moderators, column 2) — while the wage gradient does not vary with river size (columns 3–6).

Table 4: Is the response concentrated where agriculture is?

	Conflict events		Real ag. wage		Nominal wage	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Agricultural dependence						
log Upstream Precip. (t)	-0.316	-0.563	-0.085**	-0.105***	-0.069**	-0.079**
	(0.346)	(0.420)	(0.037)	(0.038)	(0.031)	(0.036)
× Agric. dependence	0.580***	0.381*	-0.066*	-0.125***	-0.017	-0.052
	(0.170)	(0.202)	(0.034)	(0.047)	(0.024)	(0.033)
log Upstream Precip. ($t-1$)	1.121***	1.022**	-0.133***	-0.150***	-0.094***	-0.103***
	(0.258)	(0.409)	(0.040)	(0.041)	(0.034)	(0.034)
× Agric. dependence	0.602***	0.748***	-0.051	-0.099***	-0.047	-0.073**
	(0.158)	(0.157)	(0.031)	(0.037)	(0.030)	(0.030)
Panel B: Agricultural dependence and the size of the river						
log Upstream Precip. (t)	-0.609	-0.663	-0.086**	-0.099**	-0.069**	-0.077**
	(0.403)	(0.412)	(0.038)	(0.040)	(0.031)	(0.036)
× Agric. dependence	1.108***	0.860**	-0.069**	-0.134***	-0.016	-0.054
	(0.330)	(0.392)	(0.034)	(0.047)	(0.024)	(0.034)
× River size	0.470	0.183	0.009	-0.012	0.003	-0.003
	(0.334)	(0.325)	(0.021)	(0.021)	(0.017)	(0.018)
× Agric. dependence × River size	0.584**	0.449*	0.004	0.012	-0.003	0.003
	(0.232)	(0.248)	(0.039)	(0.038)	(0.026)	(0.027)
log Upstream Precip. ($t-1$)	0.904***	0.882**	-0.117***	-0.129***	-0.085**	-0.090***
	(0.276)	(0.371)	(0.040)	(0.041)	(0.034)	(0.034)
× Agric. dependence	1.103***	1.284***	-0.046	-0.110**	-0.032	-0.063*
	(0.289)	(0.295)	(0.033)	(0.043)	(0.031)	(0.036)
× River size	0.064	0.288	-0.054*	-0.066**	-0.031	-0.039*
	(0.273)	(0.255)	(0.029)	(0.029)	(0.022)	(0.023)
× Agric. dependence × River size	0.371***	0.383***	-0.009	0.016	-0.033	-0.024
	(0.133)	(0.113)	(0.040)	(0.049)	(0.034)	(0.038)
Observations	6,372	6,336	4,556	4,533	4,556	4,533
Number of Districts	177	176	276	275	276	275
Mean of dependent variable	1.06	1.07				
Estimation	Poisson	Poisson	OLS	OLS	OLS	OLS
District & Year FE	✓	✓	✓	✓	✓	✓
Rival moderators × rainfall		✓		✓		✓

Notes: Every column interacts current and lagged upstream and downstream rainfall with agricultural dependence: 2005 Economic Census non-agricultural employment per head, reversed so that higher means more dependent on agriculture. Columns 1–2 are Poisson on located UCDP events, 1990–2025, with the specification of Table 2, column 4; columns 3–6 are OLS on the male field-labour wage, deflated by the local paddy price (3–4) and nominal (5–6), with the specification of Online Appendix Table D.1. Columns 2, 4 and 6 add the same interactions with three rival moderators: 2000 irrigated share, 2001 literacy rate and 2001 log worker density. Panel B adds interactions with river size (log upstream drainage area at the district's largest-drainage river cell, zero where there is none) and with river size times agricultural dependence. All moderators are standardised, pre-window cross-sections; their main effects are absorbed by district effects. Local weather enters uninteracted, and the downstream terms are not shown. Standard errors clustered by district in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Figure 6 shows when conflict responds. Replacing the annual lag with twenty-four monthly horizons, conflict is no higher during the monsoon or the kharif harvest, rises from January of $t+1$, and stays higher through that year. The kharif harvest is in by December, so January is when a lost harvest begins to cost households income: there is no crop to sell and little farm work until the next monsoon (Vanden Eynde, 2018). The rise is in battles between the state and armed groups, 66 percent of events: summed over January–June of $t+1$ their response is 4.15 ($p = 0.002$) and over July–December 3.42 ($p = 0.001$), while armed groups' attacks on civilians do not respond (Online Appendix Figure D.2). This pattern is consistent with a rise in the supply of fighters. Recruits raised when income is lost must be trained before they fight, so the extra fighting arrives over the following months rather than at once (Vanden Eynde, 2018).

VI. Infrastructure and Policy

Two policy questions follow naturally from the mechanism. If conflict responds to water the river delivers, then dams might mediate the effect. If the shock works through lost farm income, then

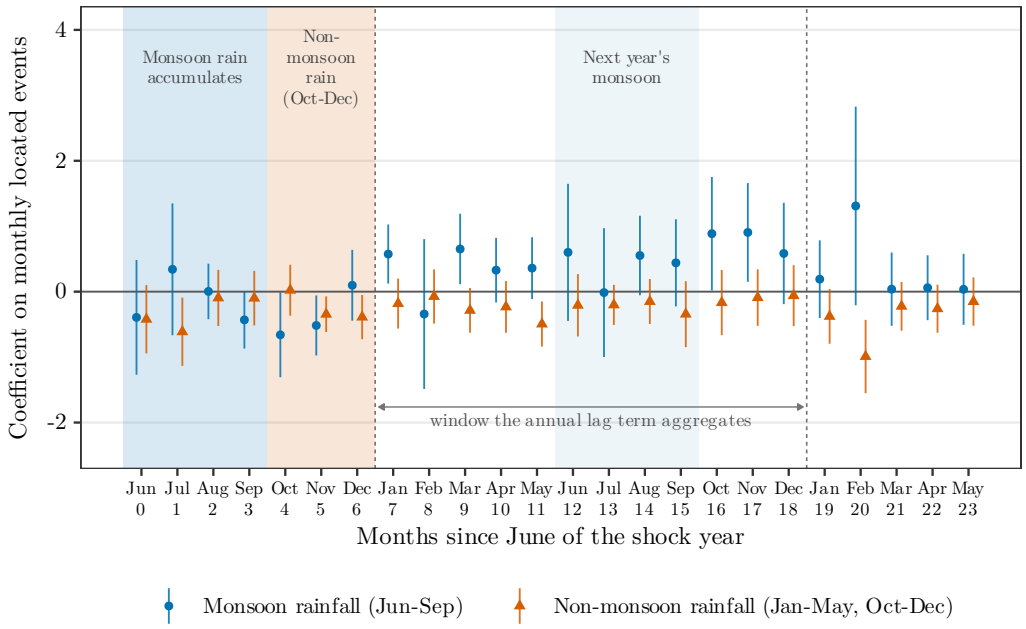


Figure 6: When conflict responds: monthly horizons of the upstream-rainfall shock

Notes: One Poisson regression of monthly located events on the monsoon (June–September) and non-monsoon (January–May, October–December) rainfall of the shock year at each of 24 monthly horizons, with district and month-year effects and the controls and spatial block of equation (1); 95 percent intervals clustered by district. Online Appendix Figure D.1 repeats the estimate without the spatial block.

India's employment guarantee might absorb it. Neither question is settled; this section reports where the evidence currently stands.

Dams.—One strategy I have tried splits each district's upstream catchment at its dams, aggregates rainfall separately over the part above them and the part below, and compare the estimated effects for the two measures. Here, the comparison is within a district, between water from two parts of the same catchment, rather than across districts with and without dams. The result does not show the pattern a buffering dam would leave: the free-flowing part (water from upstream catchment below dams) carries the conflict response, but rainfall on the regulated part arrives downstream as more flood, not less (Online Appendices E.4 and B.5; Table E.16), and the contrast rests on a single state. I am still working on this design and on others, such as instrumenting dam placement with river gradient following Duflo and Pande (2007).

Public employment.—For the rain that falls on a district's own fields, the safety net works as designed: Fetzer (2020) shows NREGA roughly halves the local rainfall–conflict relationship, his result replicates here exactly, and it survives replacing his rainfall measure with this paper's and his news-based conflict counts with UCDP (Online Appendix Table D.19). For the water that arrives from upstream, the same programme does nothing detectable. Identified from the staggered roll-out — comparing districts whose programme has already switched on with districts in the same region and year whose programme has not — the upstream–NREGA interaction is zero, and the upstream elasticity is essentially unchanged before and after (Online Appendix Table D.20). The household data agree: the workers the shock displaces do not move onto the programme, and stripping programme earnings out of the ledger barely changes the measured loss (Online Appendix D.3). A reading consistent with all three findings is administrative — relief is triggered by local rainfall deficit and drought declaration, and an upstream shock destroys a crop

while local rainfall looks unremarkable — but confirming it requires administrative programme data I have not yet obtained. For now the finding is the gap itself: the absorber that works for local rain does not reach the shock this paper studies.

VII. Conclusion

This paper asks whether rain that falls upstream raises violent conflict downstream. Monsoon rainfall over a district's drainage area — aggregated through 21,622 sub-basins of the hydrological network — raises conflict events by 17 percent per standard deviation in the following year, while the same monsoon's rain on the district itself lowers them. To build an exogenous measure of the river discharge and flooding that upstream rain causes in each district, I simulate the water's path with two standard hydrological models: one turns the catchment's rainfall into runoff, and the other carries that runoff down the river network to the district. I find that the simulated discharge raises conflict events by about 24 percent per standard deviation in the following year. Additional findings suggest that (i) the intensifying effect of upstream rainfall is not transmitted through trade networks, and (ii) the effect is concentrated in districts that depend on agriculture, follows the harvest calendar, and comes with a fall in the agricultural wage. These are consistent with the hypothesis that upstream rainfall arrives downstream as a hazard to farmers and agricultural workers and lowers the opportunity cost of joining conflict in rural India.

The paper makes two contributions. First, to the best of my knowledge, it is the first empirical study to measure rainfall over a district's entire upstream drainage area using the hydrological network, and it shows that the measure matters: in India, rain upstream and rain on the district itself push conflict in opposite directions, so a study that uses only local rainfall, as the climate–conflict literature has done, mixes the two and may understate both. Second, it is the first to replace observed flood and discharge data, which are likely to suffer from endogeneity, with an exogenous measure simulated by hydrological models.

The results point to two policy implications. First, flooding matters, not only drought. Empirical work on rainfall and conflict, including the work on India, has largely read rainfall shocks as droughts, income losses from too little rain; the shock documented here is an increase in water, arriving from upstream in each district. Second, India's employment guarantee, which mitigates the effect of local rainfall on conflict (Fetzer, 2020), does not mitigate the effect of upstream rainfall: his result replicates here, but the programme does nothing detectable to the upstream effect. Existing policy may therefore prevent only part of rainfall-driven conflict.

Two questions remain open: whether dams dampen or redirect the water's effect, and whether the employment guarantee can reach a shock that arrives from upstream, which requires administrative programme data. I will continue to pursue these questions during the PhD study.

References

Adão, R., Kolesár, M., and Morales, E. (2019). Shift-share designs: Theory and inference. *Quarterly Journal of Economics*, 134(4):1949–2010.

Bates, P. D., Horritt, M. S., and Fewtrell, T. J. (2010). A simple inertial formulation of the shallow water equations for efficient two-dimensional flood inundation modelling. *Journal of Hydrology*, 387(1–2):33–45.

Beck, H. E., Pan, M., Lin, P., Seibert, J., van Dijk, A. I. J. M., and Wood, E. F. (2020). Global fully distributed parameter regionalization based on observed streamflow from 4,229 headwater catchments. *Journal of Geophysical Research: Atmospheres*, 125(17):e2019JD031485.

Borusyak, K. and Hull, P. (2023). Nonrandom exposure to exogenous shocks. *Econometrica*, 91(6):2155–2185.

Borusyak, K., Hull, P., and Jaravel, X. (2022). Quasi-experimental shift-share research designs. *Review of Economic Studies*, 89(1):181–213.

Brakenridge, G. R. (2023). Global active archive of large flood events. Dartmouth Flood Observatory, University of Colorado.

Burke, M., Hsiang, S. M., and Miguel, E. (2015). Climate and Conflict. *Annual Review of Economics*, 7:577–617.

Castells-Quintana, D., del Pilar Lopez-Uribe, M., and McDermott, T. K. (2022). Population displacement and urban conflict: Global evidence from more than 3300 flood events. *Journal of Development Economics*, 158:102922.

Christian, P. and Barrett, C. B. (2024). Spurious regressions and panel IV estimation: Revisiting the causes of conflict. *The Economic Journal*, 134(659):1069–1099.

Crost, B., Duquennois, C., Felter, J. H., and Rees, D. I. (2018). Climate change, agricultural production and civil conflict: Evidence from the Philippines. *Journal of Environmental Economics and Management*, 88:379–395.

Dal Bó, E. and Dal Bó, P. (2011). Workers, Warriors, and Criminals: Social Conflict in General Equilibrium. *Journal of the European Economic Association*, 9(4):646–677.

Delforge, D., Wathelet, V., Below, R., Lanfredi Sofia, C., Tonnelier, M., van Loenhout, J. A. F., and Speybroeck, N. (2025). EM-DAT: The emergency events database. *International Journal of Disaster Risk Reduction*, 124:105509.

Dell, M., Jones, B. F., and Olken, B. A. (2014). What do we learn from the weather? the new climate-economy literature. *Journal of Economic Literature*, 52(3):740–798.

Dube, O. and Vargas, J. F. (2013). Commodity Price Shocks and Civil Conflict: Evidence from Colombia. *The Review of Economic Studies*, 80(4):1384–1421.

Duflo, E. and Pande, R. (2007). Dams. *Quarterly Journal of Economics*, 122(2):601–646.

Fetzer, T. (2020). Can Workfare Programs Moderate Conflict? Evidence from India. *Journal of the European Economic Association*, 18(6):3337–3375.

Funk, C., Peterson, P., Harrison, L., et al. (2026). The Climate Hazards Center infrared precipitation with stations, version 3. *Scientific Data*, 13:718.

Funk, C., Peterson, P., Peterson, S., et al. (2019). A high-resolution 1983–2016 Tmax climate data record based on infrared temperatures and stations by the Climate Hazard Center. *Journal of Climate*, 32(17):5639–5658.

Gatti, N., Baylis, K., and Crost, B. (2021). Can Irrigation Infrastructure Mitigate the Effect of Rainfall Shocks on Conflict? Evidence from Indonesia. *American Journal of Agricultural Economics*, 103(1):211–231.

Gehring, K. and Schaudt, P. (2026). Insuring peace: Index-based livestock insurance, droughts, and conflict. *The Quarterly Journal of Economics*, 141(2):1269–1334.

Goldsmith-Pinkham, P., Sorkin, I., and Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8):2586–2624.

Harari, M. and La Ferrara, E. (2018). Conflict, Climate, and Cells: A Disaggregated Analysis. *The Review of Economics and Statistics*, 100(4):594–608.

Hidalgo, F. D., Naidu, S., Nichter, S., and Richardson, N. (2010). Economic determinants of land invasions. *The Review of Economics and Statistics*, 92(3):505–523.

Hsiang, S. M., Burke, M., and Miguel, E. (2013). Quantifying the influence of climate on human conflict. *Science*, 341(6151):1235367.

Lehner, B. and Grill, G. (2013). Global river hydrography and network routing: Baseline data and new approaches to study the world's large river systems. *Hydrological Processes*, 27(15):2171–2186.

McGuirk, E. and Burke, M. (2020). The Economic Origins of Conflict in Africa. *Journal of Political Economy*, 128(10):3940–3997.

McGuirk, E. F. and Nunn, N. (2025). Transhumant pastoralism, climate change, and conflict in Africa. *The Review of Economic Studies*, 92(1):404–441.

Patel, D. (2024). Floods. Working paper, Harvard University.

Sarsons, H. (2015). Rainfall and conflict: A cautionary tale. *Journal of Development Economics*, 115:62–72.

Sundberg, R. and Melander, E. (2013). Introducing the UCDP georeferenced event dataset. *Journal of Peace Research*, 50(4):523–532.

Tellman, B., Sullivan, J. A., Kuhn, C., Kettner, A. J., Doyle, C. S., Brakenridge, G. R., Erickson, T. A., and Slayback, D. A. (2021). Satellite imaging reveals increased proportion of population exposed to floods. *Nature*, 596(7870):80–86.

Vanden Eynde, O. (2018). Targets of violence: Evidence from India's Naxalite conflict. *The Economic Journal*, 128(609):887–916.

Verdin, A., Funk, C., Peterson, P., Landsfeld, M., Tuholske, C., and Grace, K. (2020). Development and validation of the CHIRTS-daily quasi-global high-resolution daily temperature data set. *Scientific Data*, 7:303.

Yamazaki, D., de Almeida, G. A. M., and Bates, P. D. (2013). Improving computational efficiency in global river models by implementing the local inertial flow equation and a vector-based river network map. *Water Resources Research*, 49(11):7221–7235.

Yamazaki, D., Ikeshima, D., Sosa, J., Bates, P. D., Allen, G. H., and Pavelsky, T. M. (2019). MERIT Hydro: A high-resolution global hydrography map based on latest topography dataset. *Water Resources Research*, 55(6):5053–5073.

Yamazaki, D., Kanae, S., Kim, H., and Oki, T. (2011). A physically based description of floodplain inundation dynamics in a global river routing model. *Water Resources Research*, 47(4):W04501.

Yamazaki, D., Sato, T., Kanae, S., Hirabayashi, Y., and Bates, P. D. (2014). Regional flood dynamics in a bifurcating mega delta simulated in a global river model. *Geophysical Research Letters*, 41(9):3127–3135.